



北京邮电大学
Beijing University of Posts and Telecommunications

BUPT Compilers Lab3 2025Fall

Name:

Student ID:

Class Number:

Container Number:

1 Interacting Bison with Flex

The commands were used:

```
make calc
echo '1+1=' | ./calc.out
echo '2*3=' | ./calc.out
echo '5-2*2=' | ./calc.out
```

The screenshot of the compilation and execution results.

```
root@53e88d23dad3:/mnt/Workspace/lab3/calc# make calc
flex lex.l
bison -t -d syntax.y
gcc syntax.tab.c -lfl -ly -D CALC_MAIN -o calc.out
root@53e88d23dad3:/mnt/Workspace/lab3/calc# echo '1+1=' | ./calc.out
= 2
root@53e88d23dad3:/mnt/Workspace/lab3/calc# echo '2*3=' | ./calc.out
= 6
root@53e88d23dad3:/mnt/Workspace/lab3/calc# echo '5-2*2=' | ./calc.out
= 1
root@53e88d23dad3:/mnt/Workspace/lab3/calc#
```

```
python3 calc_test.py
```

```
root@53e88d23dad3:/mnt/Workspace/lab3/calc# python3 calc_test.py
pass all test cases
```

2 Validating parentheses

2.1 Code

The implementation of syntax.y is as follows.

```
%{
    #include "lex.yy.c"
    void yyerror(const char *s);
    int result;
}%

%token LP RP LB RB LC RC

%%

StringList
    : /* empty */
    | StringList String
    ;

String
    : LP StringList RP
    | LB StringList RB
```

```

    | LC StringList RC
;
%%

void yyerror(const char *s){
    result = 0;
}

int validParentheses(char *expr){
    result = 1;
    yy_scan_string(expr);
    yyparse();
    return result;
}

```

```

syntax.y
parentheses > syntax.y
1  %{
2      #include "lex.yy.c"
3      void yyerror(const char *s);
4      int result;
5  %}
6
7  %token LP RP LB RB LC RC
8
9  %%
10 StringList
11     : /* empty */
12     | StringList String
13     ;
14
15 String
16     : LP StringList RP
17     | LB StringList RB
18     | LC StringList RC
19     ;
20 %%
21
22 void yyerror(const char *s){
23     result = 0;
24 }
25
26 int validParentheses(char *expr){
27     result = 1;
28     yy_scan_string(expr);
29     yyparse();
30     return result;
31 }
32

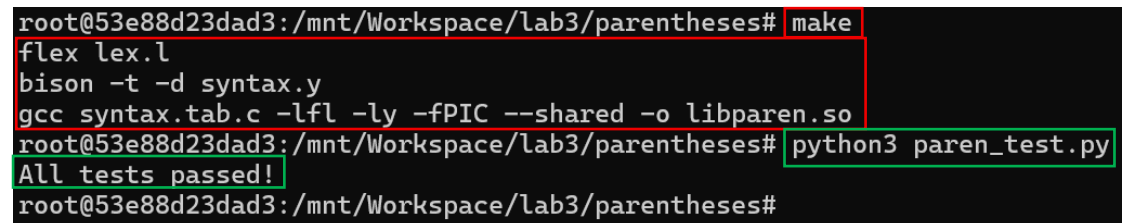
```

2.2 Run

The commands were used:

```
make
python3 paren_test.py
```

The screenshot of the compilation and execution results.

A terminal window screenshot showing the compilation and execution of a program. The prompt is root@53e88d23dad3:/mnt/Workspace/lab3/parentheses#. The commands and their outputs are: 'make' (highlighted with a red box) which runs 'flex lex.l', 'bison -t -d syntax.y', and 'gcc syntax.tab.c -lfl -ly -fPIC --shared -o libparen.so'; 'python3 paren_test.py' (highlighted with a green box) which outputs 'All tests passed!' (also highlighted with a green box). The final prompt is root@53e88d23dad3:/mnt/Workspace/lab3/parentheses#.

```
root@53e88d23dad3:/mnt/Workspace/lab3/parentheses# make
flex lex.l
bison -t -d syntax.y
gcc syntax.tab.c -lfl -ly -fPIC --shared -o libparen.so
root@53e88d23dad3:/mnt/Workspace/lab3/parentheses# python3 paren_test.py
All tests passed!
root@53e88d23dad3:/mnt/Workspace/lab3/parentheses#
```